

§ 2.5 重力场中的理想气体的性质

一、粒子数密度随高度的分布

分子在重力场中的势能

$$\varphi = mgz$$

$$\varepsilon = \varepsilon_1 + mgz$$

$$\therefore da = e^{-\alpha - \beta\varepsilon} \frac{d\omega}{h^\gamma} = \frac{N}{Z_1} e^{-\beta\varepsilon} \frac{d\omega}{h^\gamma}$$

$$= \frac{N}{Z_1 h^\gamma} e^{-\frac{mgz}{kT} - \frac{\varepsilon_1}{kT}} dx dy dz dp_x dp_y dp_z$$

$$da_z = \frac{N}{Z_1 h^\gamma} e^{-\frac{mgz}{kT}} dz \int e^{-\frac{\varepsilon_1}{kT}} dx dy dp_x dp_y dp_z$$

$$Z_1 = \int e^{-\frac{mgz}{kT}} dz \int e^{-\frac{\varepsilon_1}{kT}} dx dy dp_x dp_y dp_z$$

$$da_z = \frac{Ne^{-\frac{mgz}{kT}} dz}{\int e^{-\frac{mgz}{kT}} dz}$$

定义：高度 z 处的分子数密度

$$n(z) = \frac{da_z}{Sdz}$$

$$\therefore n(z) = \frac{Ne^{-\frac{mgz}{RT}}}{S \int e^{-\frac{mgz}{RT}} dz} = n(0)e^{-\frac{mgz}{RT}}$$

二、重力场中压强随高度的变化

对于理想气体： $pV = NkT$

$$p(z) = n(z)kT = n(0)kTe^{-\frac{mgz}{RT}} = p(0)e^{-\frac{mgz}{RT}}$$

三、重力场中的内能和热容量

设气体高度为 H ，截面积为 S

$$\begin{aligned} Z_1 &= \frac{1}{h^\gamma} \int e^{-\frac{mgz}{kT}} dz \int e^{-\frac{\varepsilon_1}{kT}} dx dy dp_x dp_y dp_z \\ &= \frac{1}{h^\gamma} \int e^{-\frac{mgz}{kT}} dz \int dx dy \int e^{-\frac{\varepsilon_1}{kT}} dp_x dp_y dp_z \end{aligned}$$

$$\begin{aligned}
 &= \frac{S}{h^3} \int e^{-\frac{\beta}{2m}(p_x^2 + p_y^2 + p_z^2)} dp_x dp_y dp_z \int_{H_0}^{H_0+H} e^{-\frac{mgz}{kT}} dz \\
 &= S \left(\frac{2\pi m}{h^2 \beta} \right)^{3/2} \frac{1}{mg\beta} \left(e^{-\beta mgz} \right) \Big|_{H_0+H}^{H_0} \\
 &= S \left(\frac{2\pi m}{h^2 \beta} \right)^{3/2} \frac{1}{mg\beta} e^{-\beta mgH_0} (1 - e^{-\beta mgH})
 \end{aligned}$$

$$\therefore U = -N \frac{\partial}{\partial \beta} \ln Z_1 = U_0 + NkT - \frac{NmgH}{e^{-\frac{mgz}{RT}} - 1}$$

$$C_V = C_V^0 + Nk - \frac{N(mgH)^2}{(e^{-\frac{mgz}{RT}} - 1)^2} \frac{1}{kT}$$